

BIKI DUTTA

Data Quality Engineer | Computer Vision | Model Training & Inference | AI/ML QA

Jorhat, Assam, India | +91 6003359148 | bikidutta319@gmail.com

[LinkedIn: linkedin.com/in/bikidutta](https://www.linkedin.com/in/bikidutta) | [Portfolio: biki97.github.io](https://biki97.github.io) | [GitHub: github.com/biki97](https://github.com/biki97)

PROFESSIONAL SUMMARY

Data Quality Engineer with end-to-end exposure to the computer vision pipeline — training image classification models (CNNs, ResNet, EfficientNet) on platforms like Roboflow, running inference on production datasets, evaluating model outputs, and feeding insights back into iterative training cycles. Combines hands-on ML workflow experience with strong software engineering fundamentals (Java, Spring Boot, Python, SQL, AWS S3) and a Master of Computer Applications (MCA) from Jorhat Engineering College. Skilled at identifying failure modes, performing root-cause analysis on misclassifications, and translating real-world data issues into actionable model improvements. Actively building deeper Python and deep-learning expertise to grow into a Machine Learning Engineer role.

CORE COMPETENCIES

- **ML Training & Inference:** Image Classification (CNNs, ResNet, EfficientNet), Object Detection (YOLO), Model Training on Roboflow, Inference on Production Datasets, Iterative Re-training
- **Computer Vision:** Multi-Class Image Classification, Bounding Box Annotation, Class Confusion Analysis, False Positive / False Negative Investigation
- **ML Evaluation:** Precision, Recall, F1 Score, Confusion Matrix (Macro / Micro / Weighted), mAP, IoU, Confidence Threshold Tuning
- **Programming:** Python, Java, SQL, JavaScript, HTML, CSS
- **ML Platforms & Tools:** Roboflow, CVAT, training-script modification, dataset versioning
- **Cloud & Infrastructure:** AWS S3 (CLI dataset management, large-scale data ops), Linux/Unix Command Line, Git, GitHub
- **Databases:** PostgreSQL, MySQL
- **Soft Skills:** Root Cause Analysis, Technical Documentation, Cross-Functional Collaboration, Detailed Status Reporting

PROFESSIONAL EXPERIENCE

Data Quality Engineer | *Borde, Inc.* — San Ramon, CA (Remote)

Feb 2026 – Present

- Own end-to-end computer vision data workflow for food-tech clients: dataset preparation, image annotation, model training, inference, evaluation, and feedback loops.
- Train and re-train image classification models (CNN-based architectures including ResNet and EfficientNet variants) on Roboflow, modifying existing training scripts to tune hyperparameters and class-balancing strategies for new datasets (e.g., Walnut GD, Diamond Foods).
- Run inference on large production datasets, then systematically evaluate predictions — distinguishing true positives from false positives, identifying confusion patterns (e.g., Light_Amber vs FM_Shell, Halves vs Pieces, background stains misclassified as objects).
- Conduct root-cause analysis on model failures — separating data-distribution shifts (new client data) from class-imbalance effects (rare classes like Red_Pieces) from visually-similar-class confusions — and feed structured findings back into the next training cycle.
- Annotate high-volume image datasets (typically 200–700 labels per day across 10+ classes including FM_Shell, FM_Fins_Attached, FM_Rocks_Dirtballs, Amber_Pieces, Light_Pieces) to support model improvement and address recurring failure modes.
- Manage AWS S3 datasets at scale using CLI workflows — listing, filtering, counting, and consolidating multi-folder reports into unified, easily-shareable structures.

- Author daily and weekly status reports summarizing annotation throughput, model behavior, observed symptoms, root causes, and proposed next steps for the engineering team.

Software Development Intern | *National Informatics Centre (NIC) — Shillong* Jan 2024 – Jun 2024

- Built backend modules for an enterprise E-VAT Tax Filing System using Core Java, Spring Boot, and PostgreSQL — building strong foundations in software engineering, data modeling, and validation.
- Designed and implemented secure REST APIs with role-based authentication for vendors, officers, and administrators.
- Optimized PostgreSQL queries and database schema to improve performance and scalability of the system.
- Followed Git-based collaboration workflows and wrote technical documentation for handover and onboarding.

Data Entry Operator – Team Leader | *Inland Water Transport (IWT), Assam* 2021 – 2022

- Led a small team of data entry operators handling government records — coordinated daily workflows, reviewed entries for accuracy, and maintained data integrity across digital and physical records.
- Built early data-quality discipline by standardizing validation steps and cross-checking entries — a foundation that directly translated into the structured data-quality work performed later at Borde.

Data Entry Operator | *Jorhat Medical College, Assam* 2020 – 2021

- Performed high-accuracy digitization of medical and administrative records, maintaining strict confidentiality and consistency standards.
- Developed strong attention-to-detail, structured documentation habits, and an ability to work with large repetitive datasets — skills now directly applied to AI/ML data work.

PROJECTS

E-VAT Tax Filing System (*Spring Boot, PostgreSQL, Thymeleaf*)

- Designed a secure, role-based filing system with vendor submission, officer review, and structured correction workflow — emphasizing data validation and integrity.
- Reduced paperwork and improved communication in the VAT process through digital approval workflows.

E-Commerce Android App (*Flutter, Dart, Firebase*)

- Built a cross-platform shopping app with product browsing, cart, secure Firebase Authentication, and real-time data sync via Firestore.

Voting Management System (*PHP, MySQL, HTML, CSS*)

- Developed a secure college election platform with separate voter/admin interfaces, session-based authentication, and transparent result reporting.

EDUCATION

Master of Computer Applications (MCA) Oct 2022 – Jul 2024

Jorhat Engineering College, Assam | CGPA: 7.08 | Cleared JMEE entrance examination

Bachelor of Science (B.Sc.) in Information Technology Jun 2015 – Jul 2018

Jorhat Institute of Science & Technology | Percentage: 63%

ADDITIONAL INFORMATION

- **Languages:** Assamese (Native), Hindi (Fluent), English (Fluent)
- **Currently Learning:** Python ML libraries (NumPy, Pandas, scikit-learn), PyTorch fundamentals, hands-on YOLOv8 training from scratch, MLOps basics
- **Career Goal:** Grow from Data Quality Engineer into a Machine Learning Engineer role, combining hands-on model-training experience with deeper Python and deep-learning expertise.